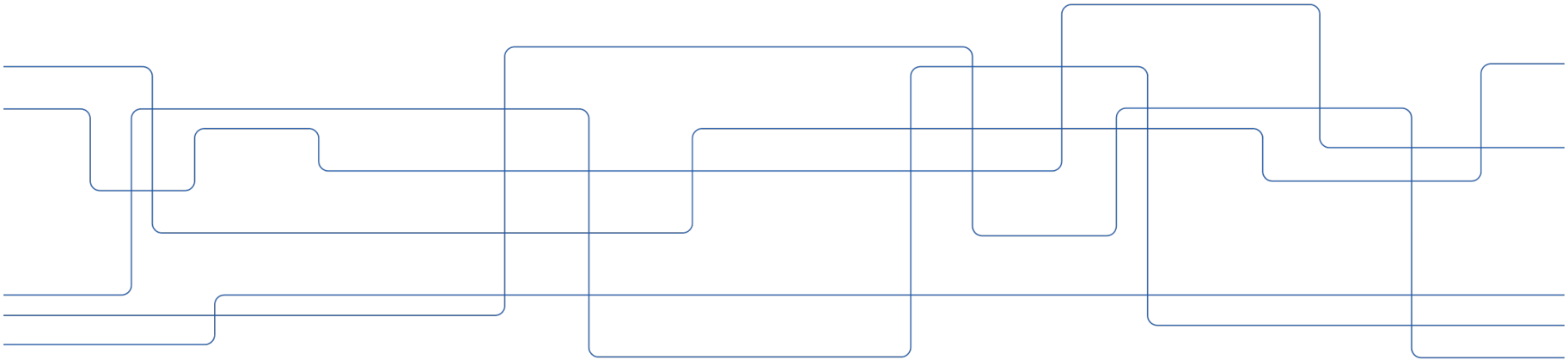
KTH ROYAL INSTITUTE
OF TECHNOLOGY

Ultrasound image simulation

Matilda Larsson

CM2014



Lecture outline

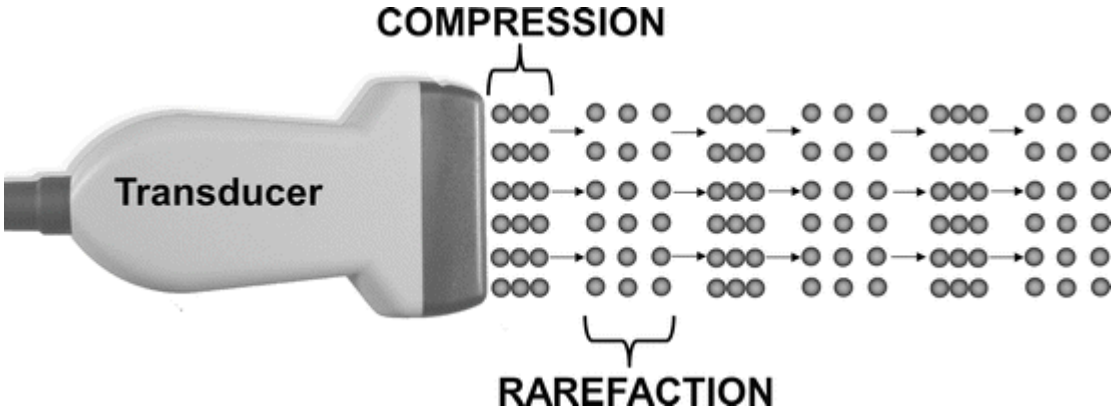
1. Ultrasound physics introduction

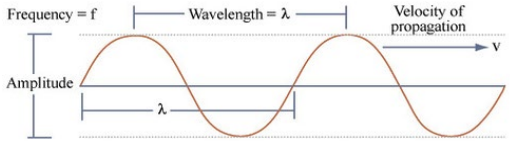
2. Ultrasound image simulation

3. Computer lab assignment

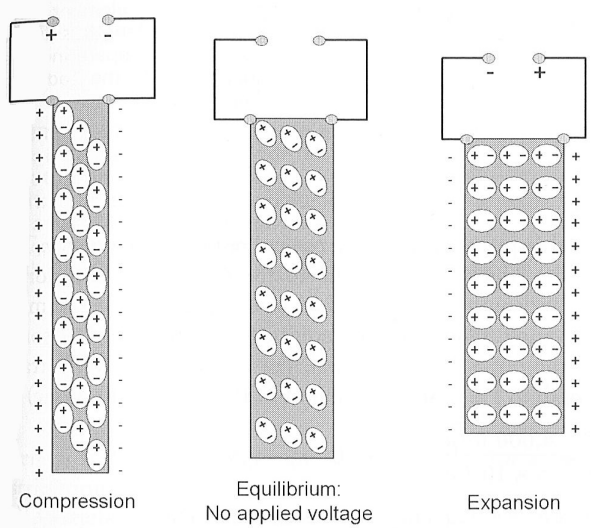
Ultrasound

- Sound at frequencies greater than 20 kHz





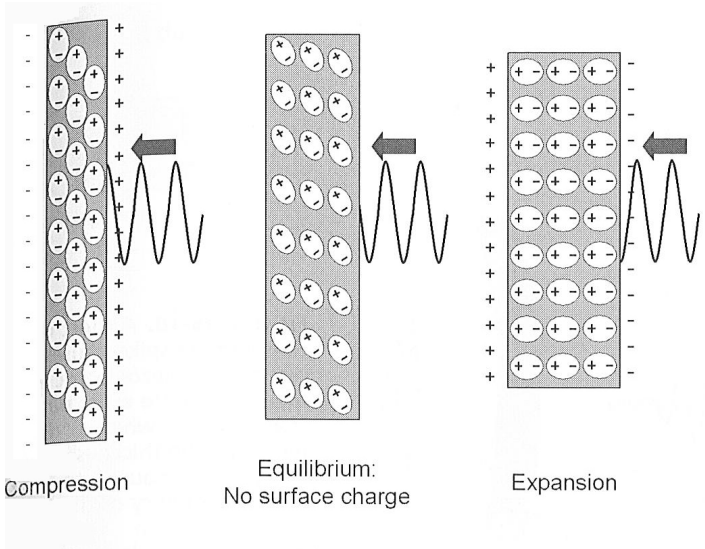
Ultrasound transmit

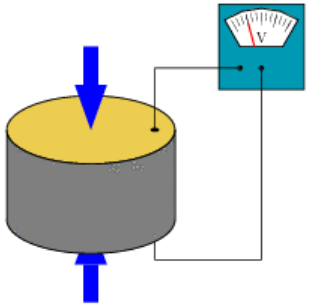


Voltage → Vibrations → Ultrasound generation

PZT - lead zirconium titanate

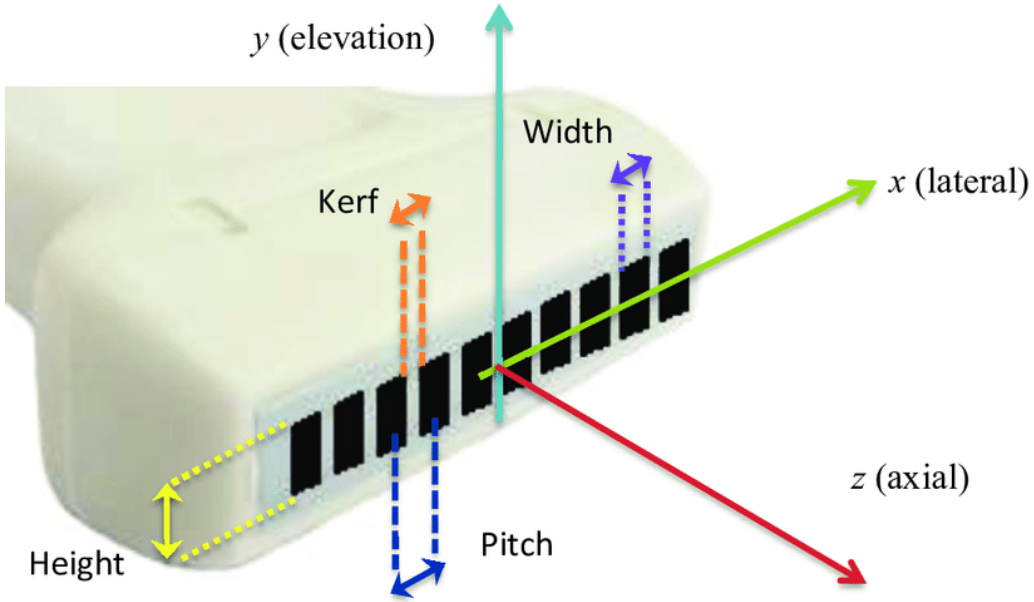
Ultrasound receive





Echoes → Vibrations → Voltage generation

Ultrasound transducer



Pulse-echo principle



Ultrasound pulse (transmit)

Tissues

Tissue interaction

- **Reflection**
- **Scattering**

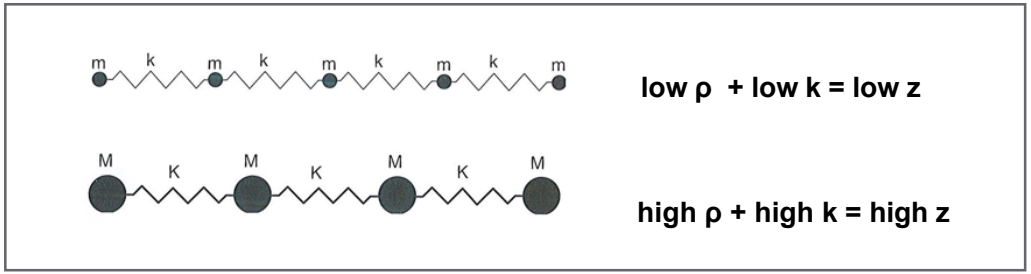
Velocity of sound in tissue

1540 m/s

To position echoes in image (in depth)

Acoustic impedance

- The resistance to the propagation of ultrasound waves through tissues
- Each tissue type has a unique acoustic impedance



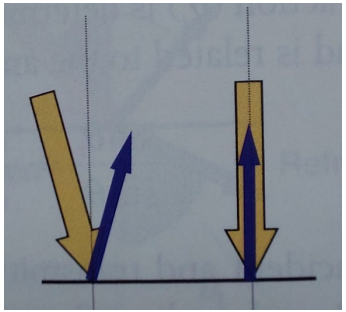
$$Z = \rho c \qquad (kg/m^2 s)$$

ρ - Density, c - Speed of sound

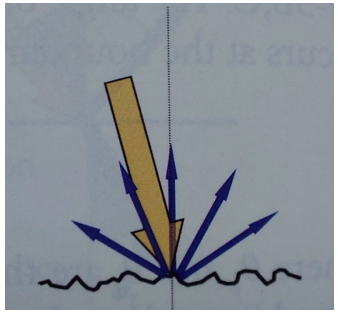
Reflection

- Wave propagation between two tissues of different acoustic impedances
- A fraction of the wave intensity is reflected back (= echo):

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$$

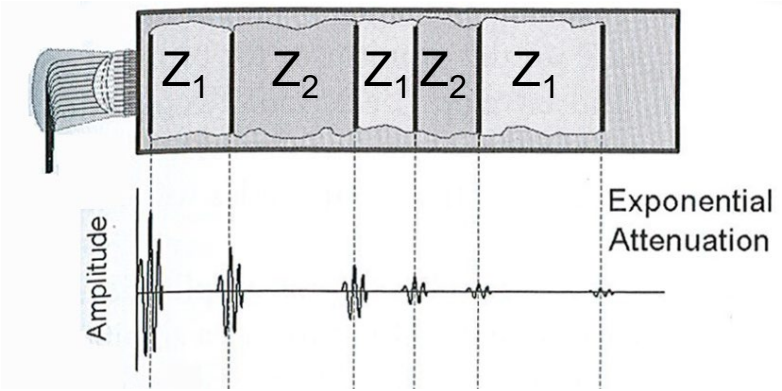


Specular (smooth) reflection



Non-specular (diffuse) reflection

Reflection and attenuation



Echo (RF-signal)

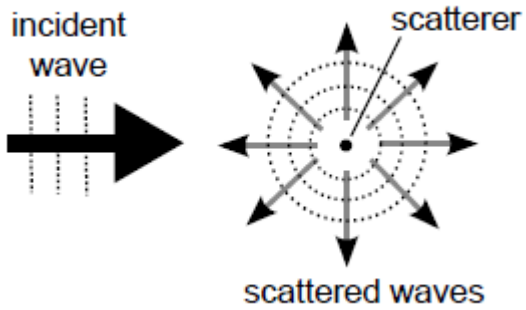
Time-gain-compensation
- Compensate for attenuation

Z, acoustic impedance

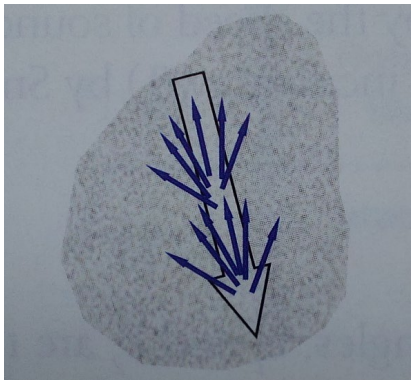
Scattering

- Ultrasound strikes a structure which

1) Has a different acoustic impedance to the surrounding tissue, and
2) Is smaller than the wavelength of the incident sound



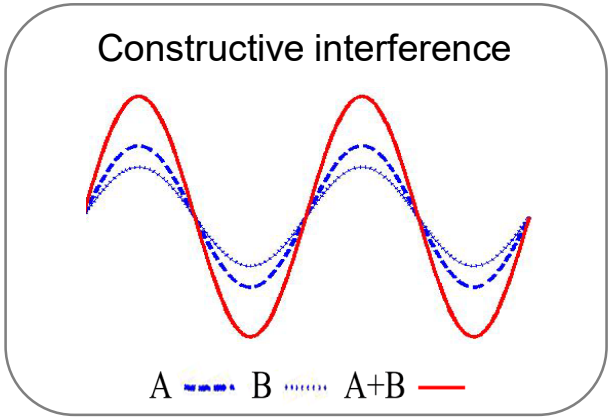
Small object reflectors
with size ≤ λ

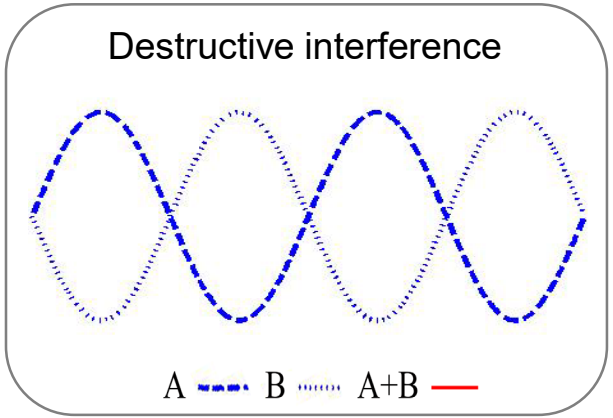


Backscattering

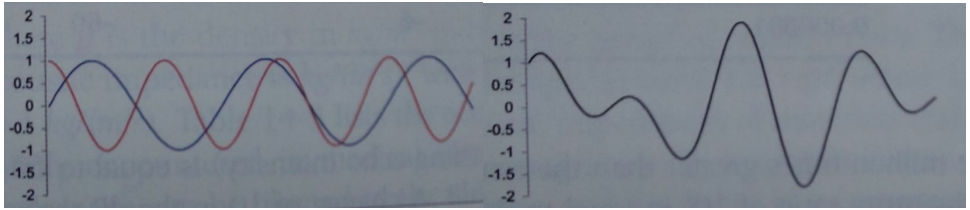
Interference

- Superposition of two or several sound waves



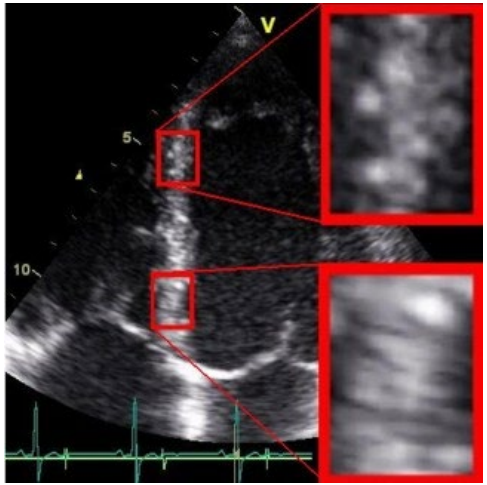


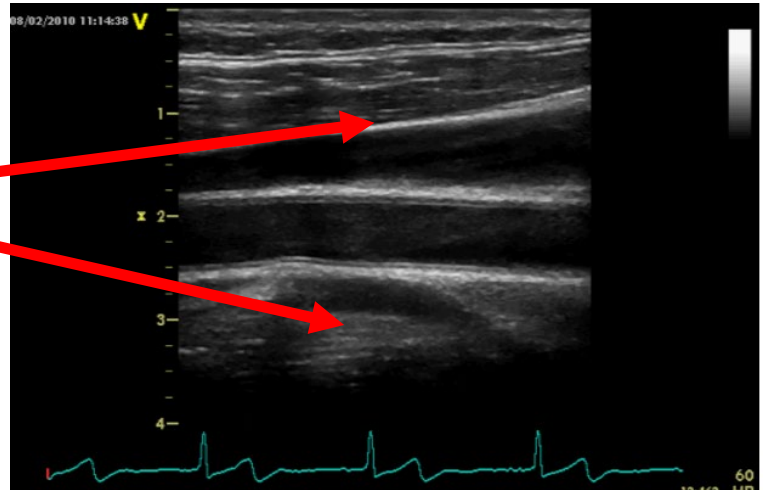
Complex interference



Speckle pattern

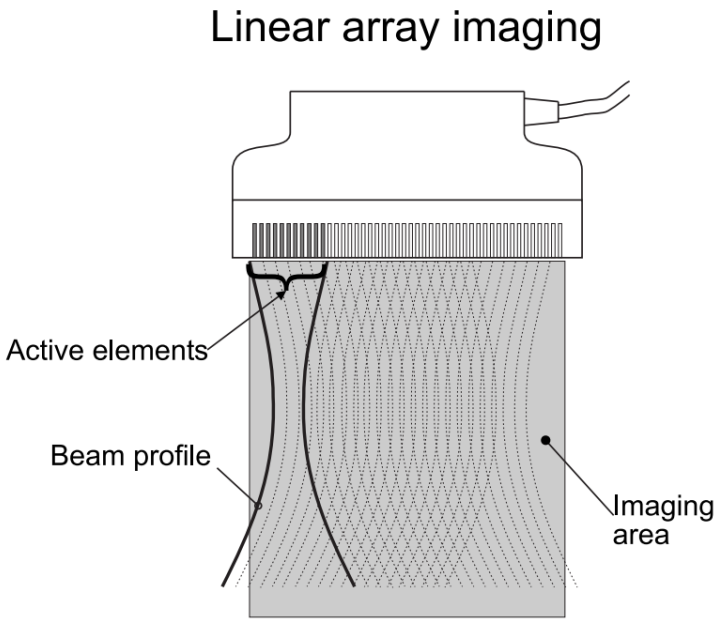
- Grayscale pattern / intensity variation in the image not revealing physical structure
- Occurs due to scattering and interference
- Fingerprint – consistent speckle pattern when imaging the same stationary structure





Scanning

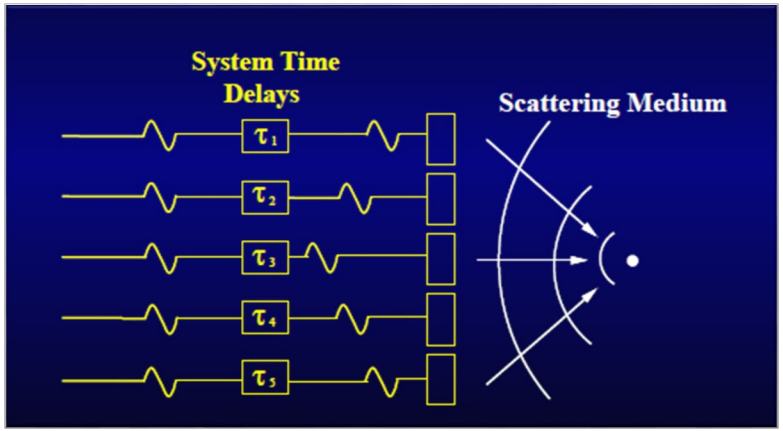
- Group-wise activation of elements for one imaging line



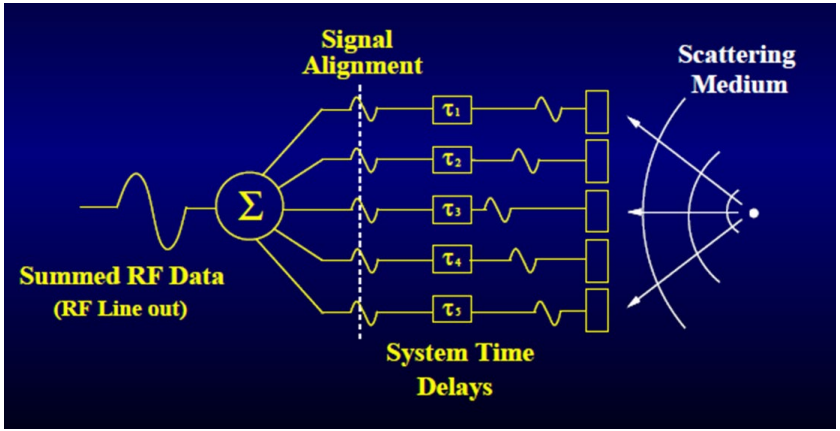
Beamforming

- Forming an interference pattern with amplification in one distinct direction - focusing
- Time delays in transmit and receive

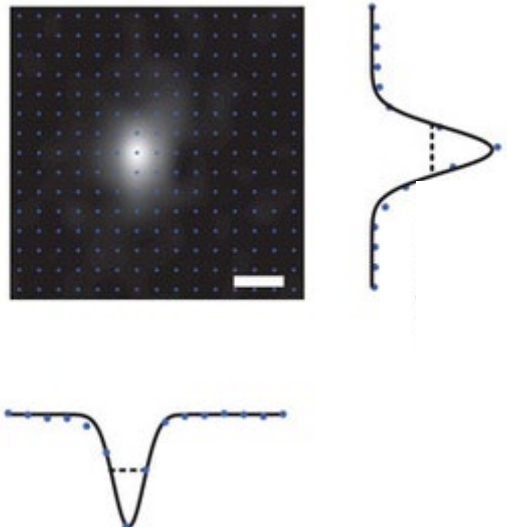
Transmit



Receive

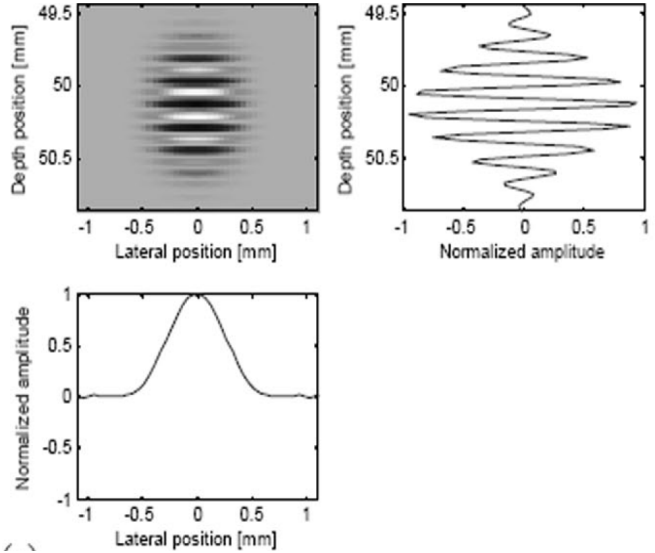


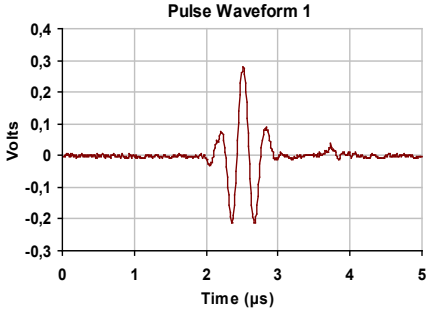
Point-spread function



Axial direction (depth)

Lateral direction (width)

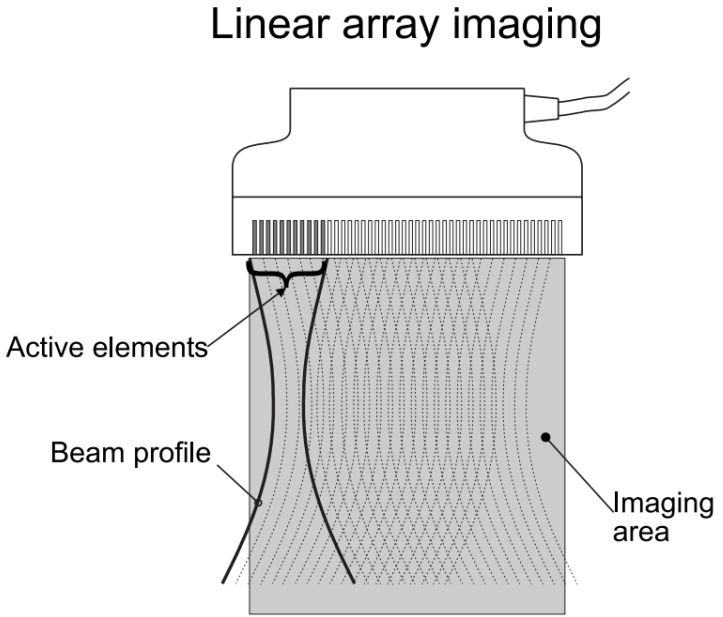




3 cycles (3MHz) pulse

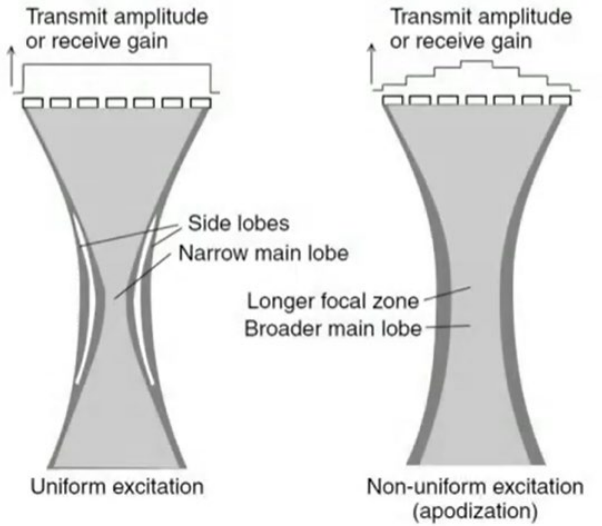
Point-spread function

- Lateral PSF varies with depth (pulse width)
- Axial PSF does not vary with depth (pulse length)



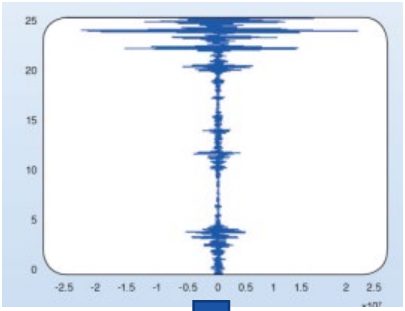
Apodization

- Vary transmit amplitude / receive gain over the aperture
- Multiplying signals in transmit and receive channels by weights
- Reducing side lobes in the image

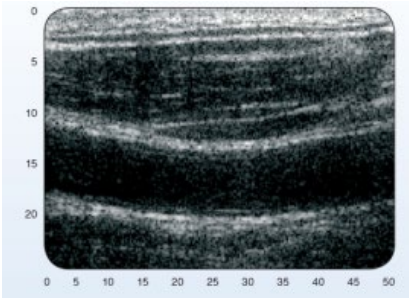


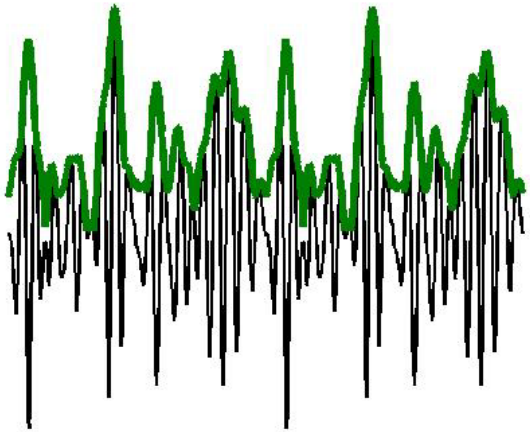
B-mode image formation

RF-signal

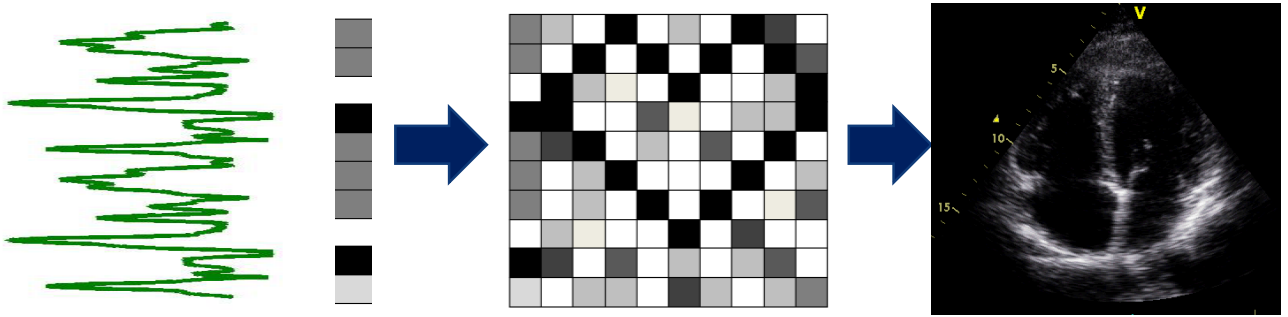


B-mode



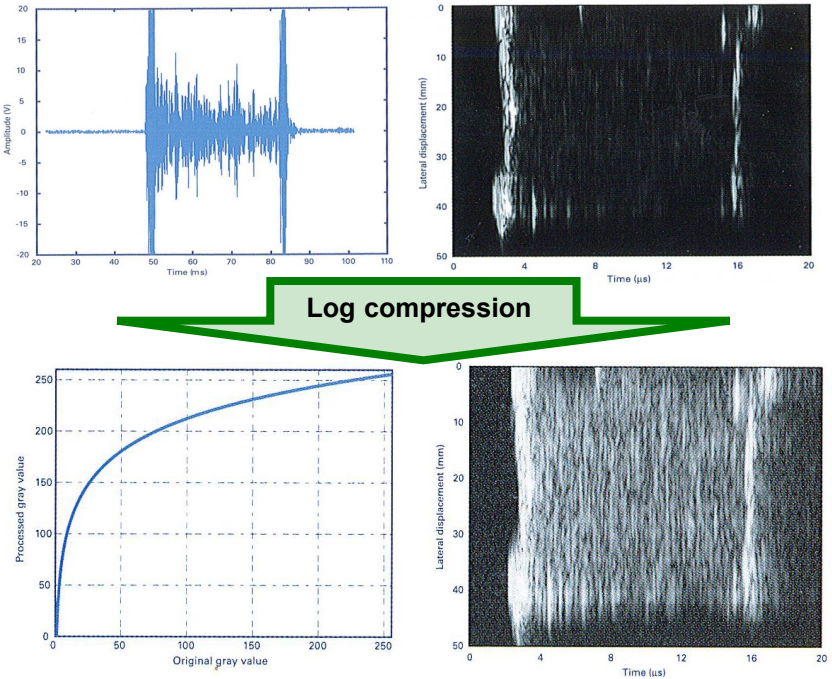


- Envelope detection (green)
  - Hilbert transform



Log compression

- Reduce the dynamic range to improve image visualization
- Specular reflections weaker + Backscattered signal stronger



Lecture outline

1. ~~Ultrasound physics introduction~~

2. **Ultrasound image simulation**

3. Computer lab assignment

Ultrasound simulation examples

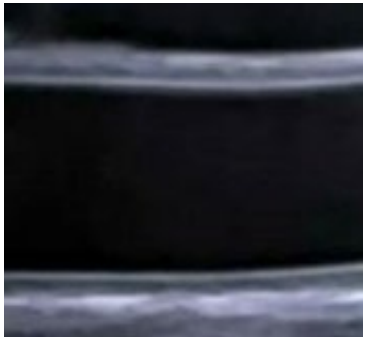
Simulation



Fetus



Left ventricle

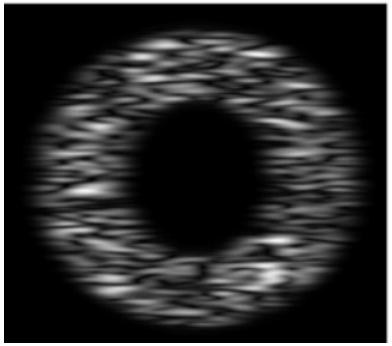


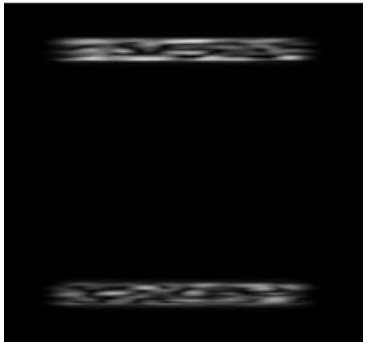
Carotid artery

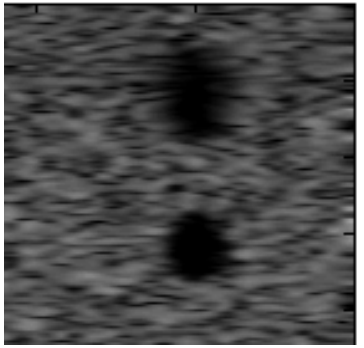


Cyst









The use of ultrasound simulations

1) Designing and evaluating ultrasound transducers, beam formers and systems
2) Image quality assessment

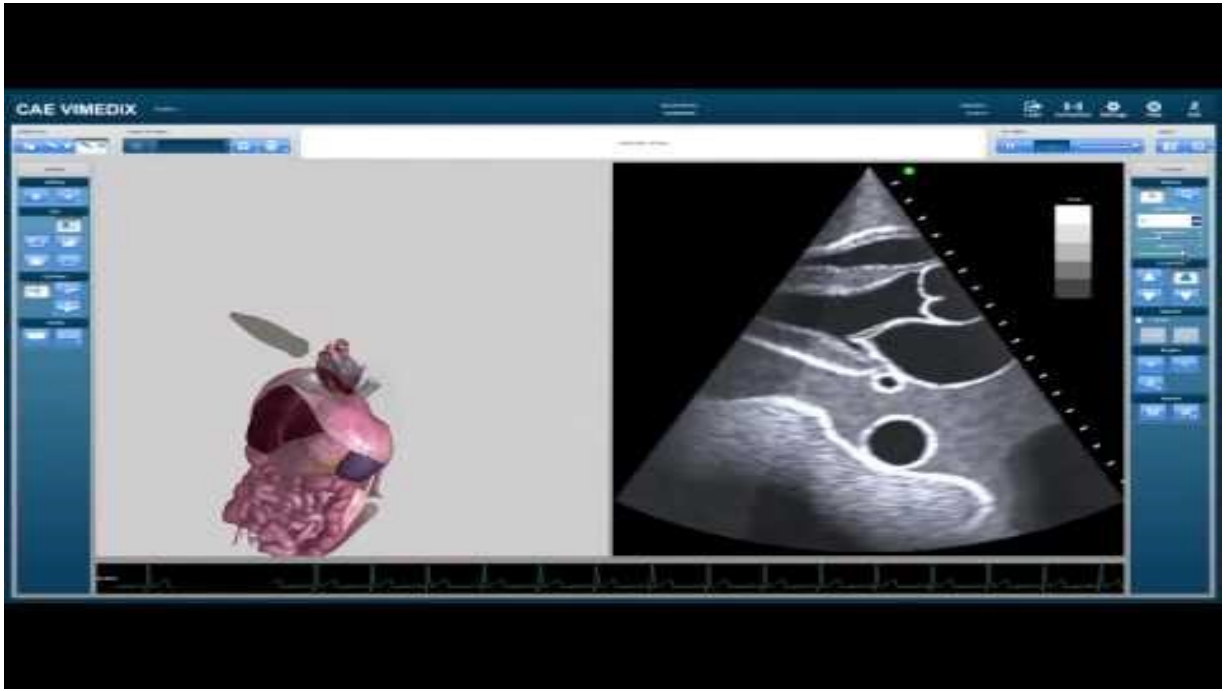
3) Flow assessment validation
4) Tissue assessment validation



Known ground truth

5) Medical training

Ultrasound simulation for medical training



https://youtu.be/6sH1usoEX70

Ultrasound simulation software

- **Field II (Jørgen Arendt Jensen, DTU) - Matlab**

Spatial impulse response method

*J.A. Jensen: Field: A Program for Simulating Ultrasound Systems, Medical & Biological Engineering & Computing, pp. 351-353, Volume 34, Supplement 1, Part 1, **1996***

- **COLE (Hang Goa, KU Leuven) - Matlab**

Fast convolution-based methodology by multiple 1-D convolutions

*Gao H et al. A fast convolution-based methodology to simulate 2-D/3-D cardiac ultrasound images. IEEE Trans Ultrason Ferroelectr Freq Control. **2009** Feb;56(2):404-9*

- **K-Wave - Matlab**

Time-domain model of acoustic wave propagation

*B. E. Treeby and B. T. Cox, "k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave-fields," J. Biomed. Opt., vol. 15, no. 2, p. 021314, **2010***



Jørgen Arendt Jensen
DTU - Copenhagen

Field II - Overview

- Simulating transmitting and receiving waveforms and the scattering mechanisms in the tissue

Step 1
- Create imaging phantom

Step 2
- Configure imaging setup

Step 3
- Simulate received waveform

Step 4
- Image generation

Step 1 – Create imaging phantom

1. Define:

- Phantom geometry
- Phantom movement (if any)
- Gray-scale intensity map (bitmap – acoustic impedance differences)

Based on imaging or FE simulations

2. Distribute initial point scatterers

- Scatter density > 10 000 / cm^3 for fully developed speckles

- Single point scatterers with high amplitude (if any)

- Scatterers for specular reflections (if any)

- Remove scatterers in regions with fluid (low/no backscattering)

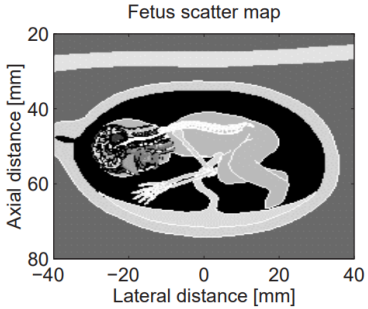
3. Set scatterers amplitude

- Randomly distributed (Gaussian distribution), scaled according to gray-scale intensity map

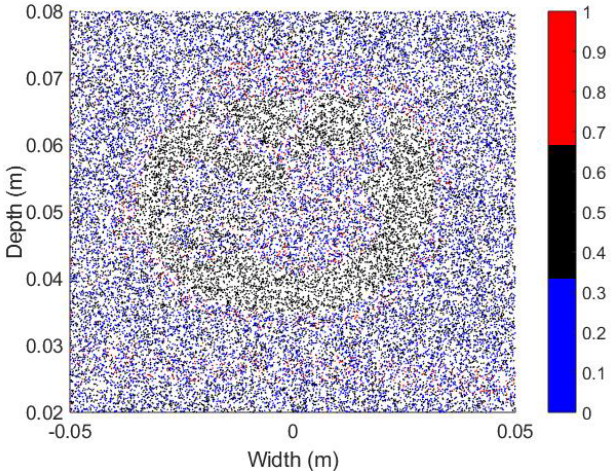
4. Move scatterers

- With frame rate as in intended imaging setup

Example: Fetus phantom

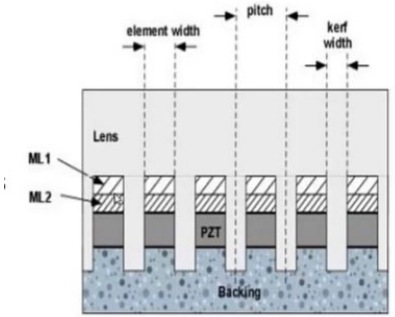


- A fetus in the third month of development
- Bitmap created based on 2D ultrasound image (static)
- 200,000 scatterers randomly distributed with amplitude based on bitmap





Step 2 – Configure imaging setup



- Transducer geometry (e.g. number and size of elements)
- Speed of sound, sampling rate etc.

Configure transmit array

- Impulse response of your transducer (natural vibration of transducer)
- Excitation (how you stimulate the transducer to create that waveform

Configure receive array

- Impulse response (probably same as transmit)

- Focus settings, beamforming, apodization etc.

Step 3 - Simulate received waveform

$$p_r(\vec{r},t) = v_{pe}(t) \underset{t}{\star} f_m(\vec{r}) \underset{r}{\star} h_{pe}(\vec{r},t)$$

p_r = received signal from the transducer

v_{pe} = pulse-echo impulse (transducer excitation and the electro-mechanical impulse response during transmit and receive)

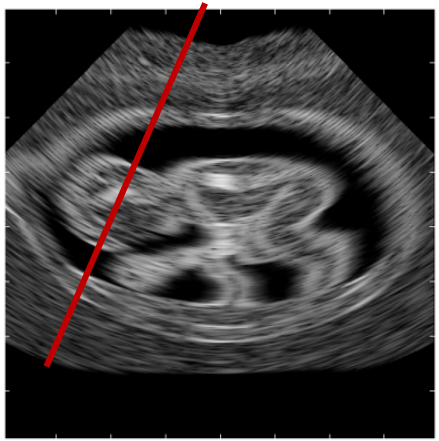
f_m = scatterers amplitude

h_{pe} = pulse-echo spatial impulse response relating transducer geometry to the spatial extent of scatterers

- Repeat for every scan line

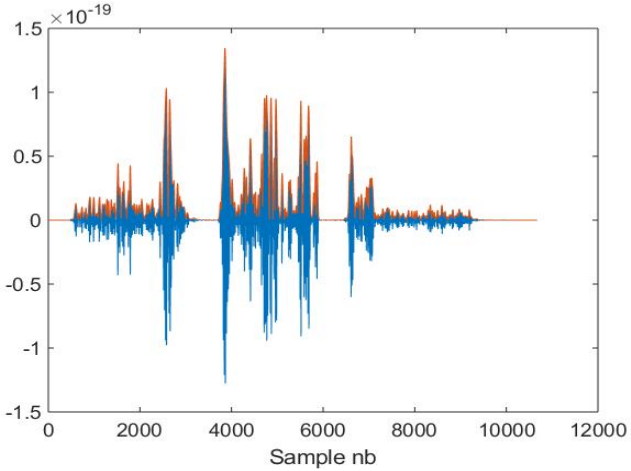
Step 4 - Image generation

Scan line 30



Scan line 30

RF-signal (blue)/Envelope-detected signal (orange)

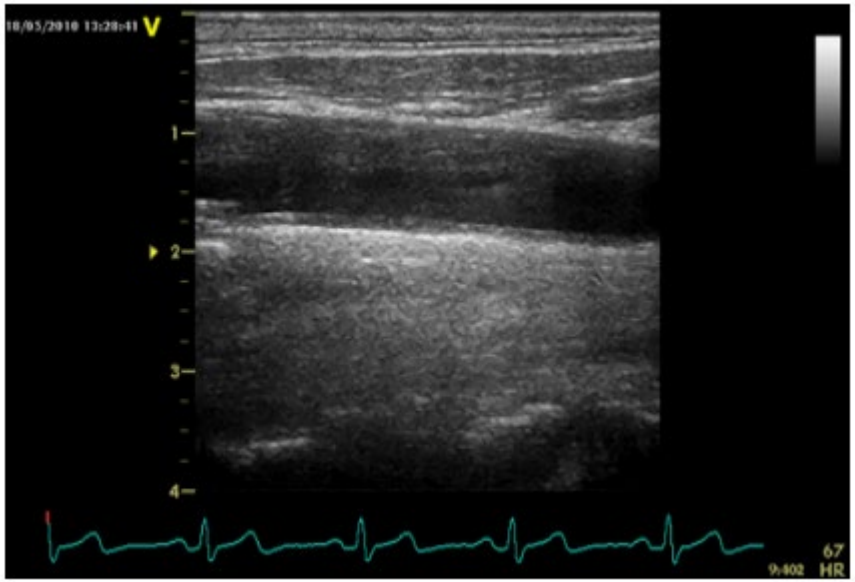


For every scan line:

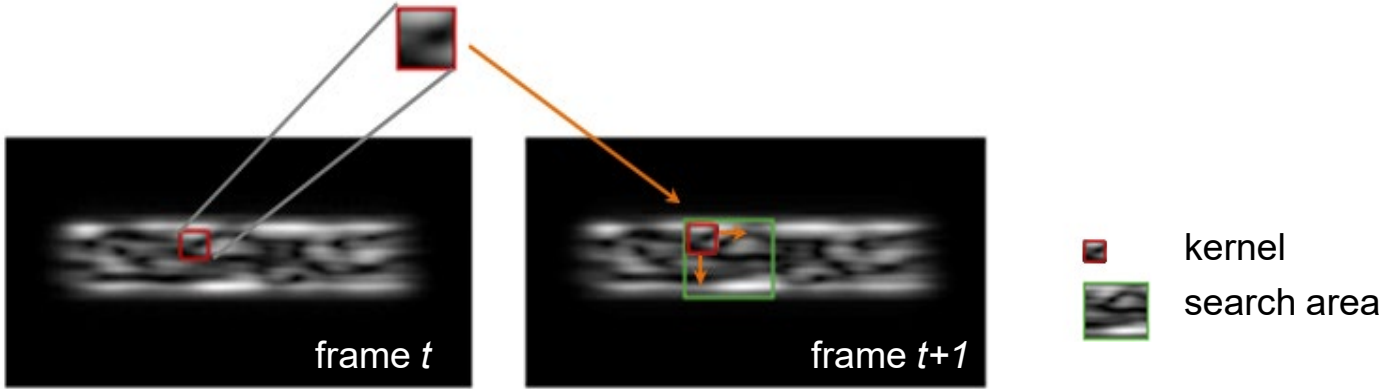
RF-signal → Envelope detection + Log compression → B-mode

Example from research

- Validation of arterial strain estimation using ultrasound speckle tracking
- Challenging because of small artery dimensions and movements

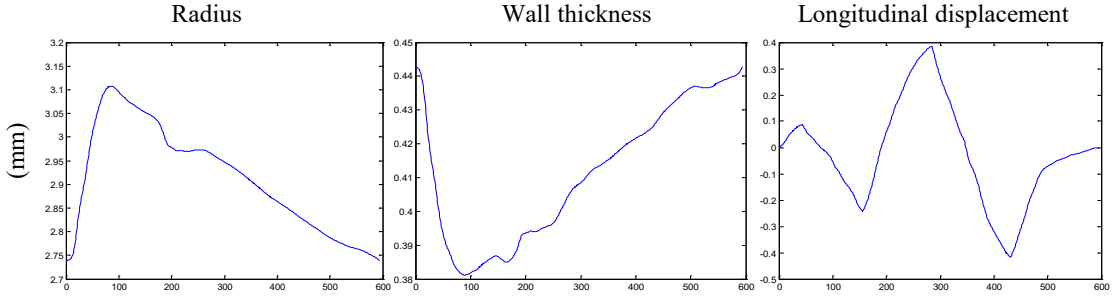


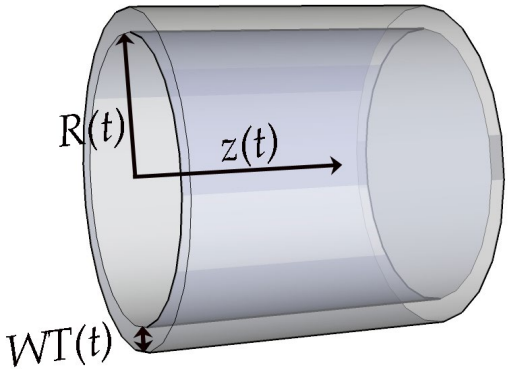
Speckle tracking for arterial strain assessment



- Block-matching using normalized cross-correlation

Carotid artery kinematic model



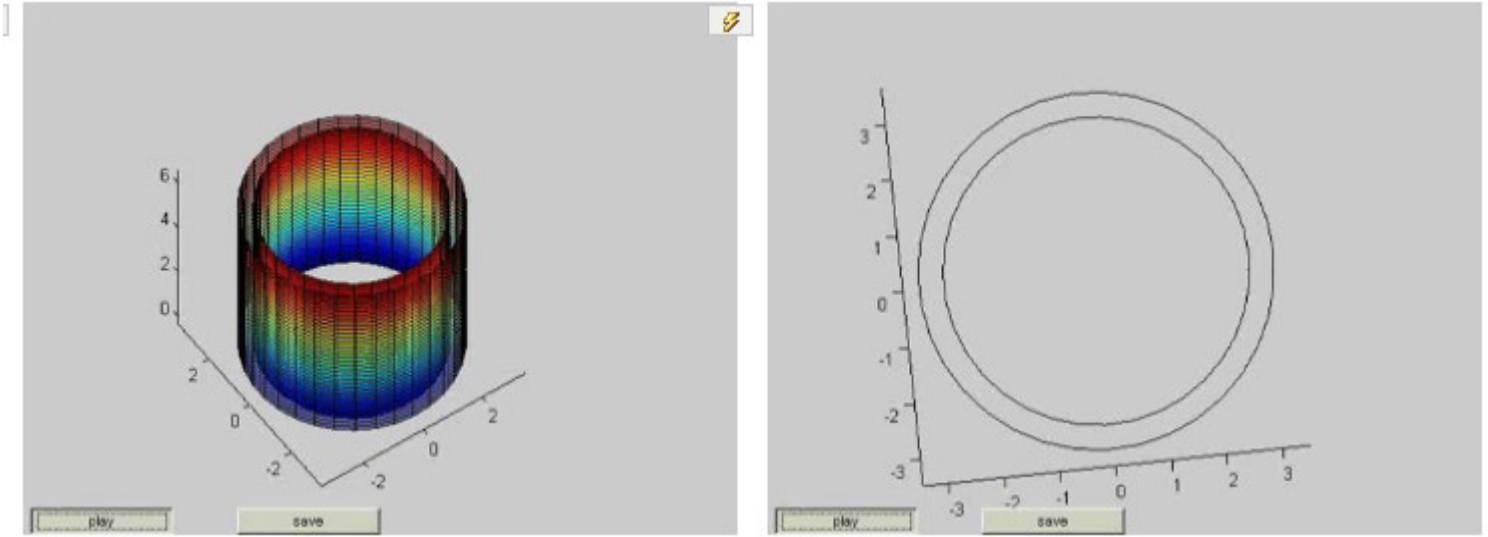


$$r(t) = \frac{WT(t)}{WT(0)}\,[r(0) - R(0)] + R(t)$$

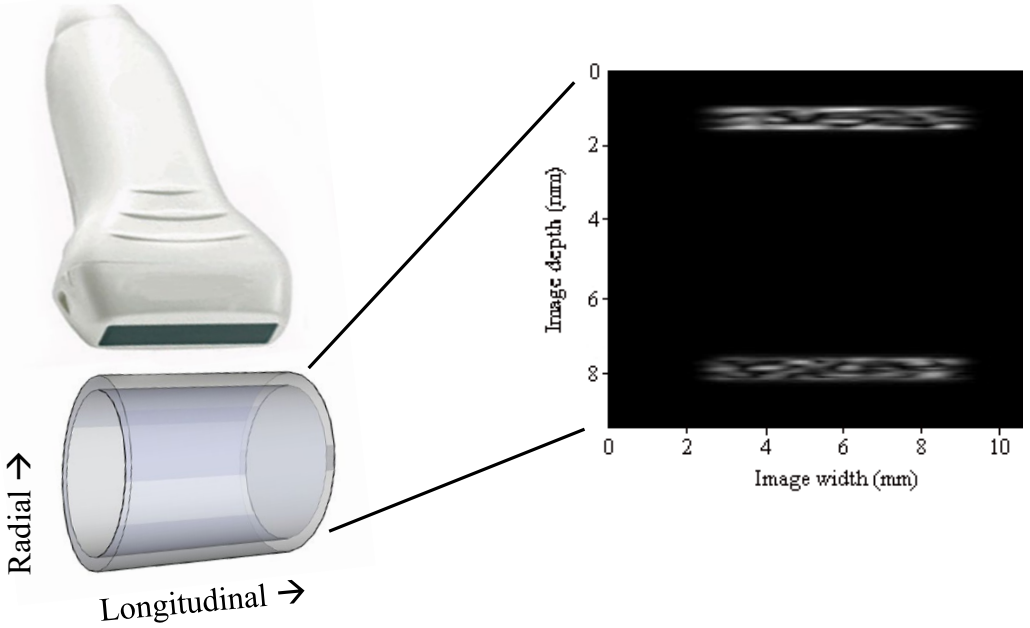
$$z(t) = z(0)\,\frac{2R(0)WT(0) + WT(0)^2}{2R(t)WT(t) + WT(t)^2} + D_{long}$$

R(t) = inner radius
z(t) = coordinate in longitudinal direction of the vessel
WT(t) = wall thickness
D_{long} = superimposed longitudinal displacement

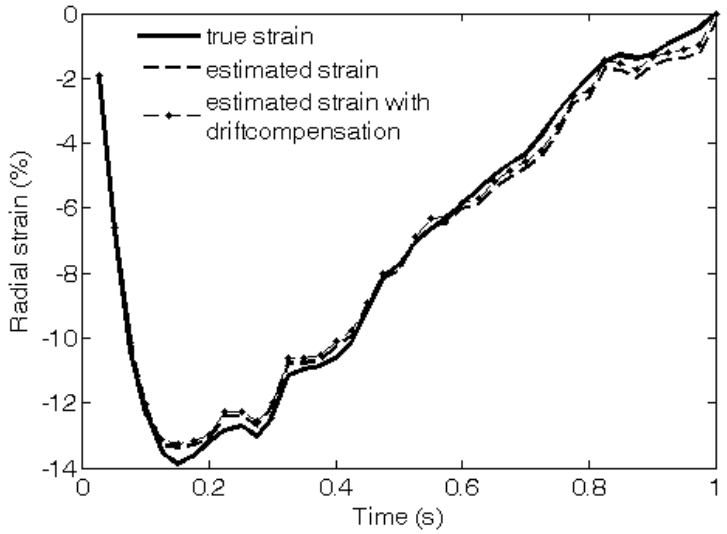
Carotid artery kinematic model

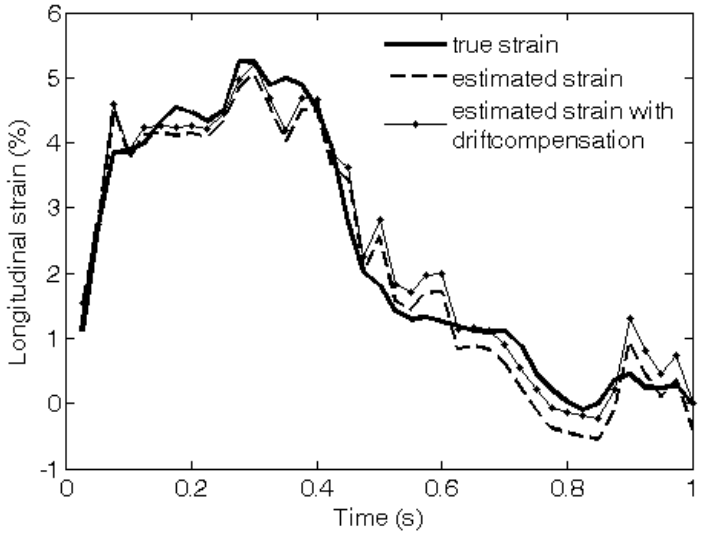


Long axis images of the model



Strain results





Lecture outline

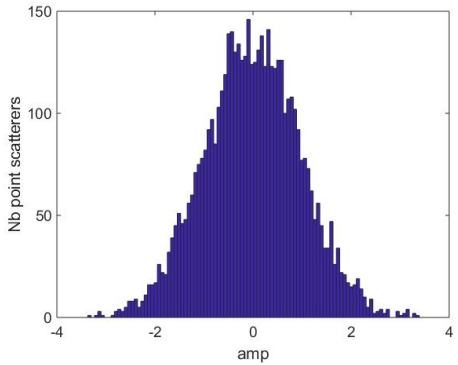
1. ~~Ultrasound physics introduction~~

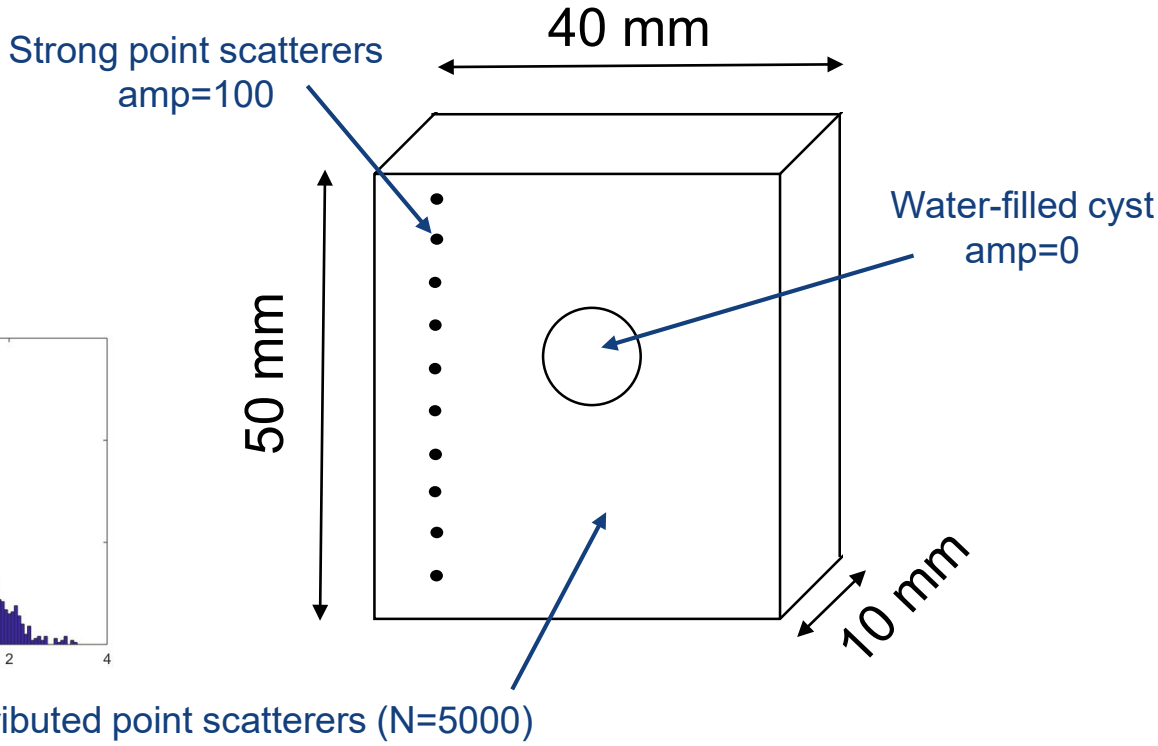
2. ~~Ultrasound image simulation~~

3. **Computer lab assignment**

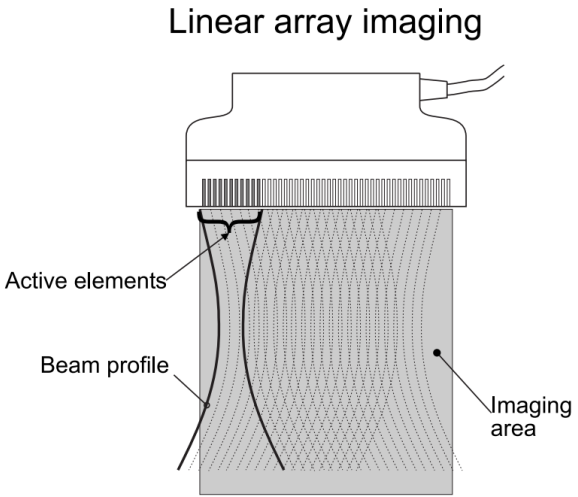
Computer Lab – Phantom

- Static cyst phantom



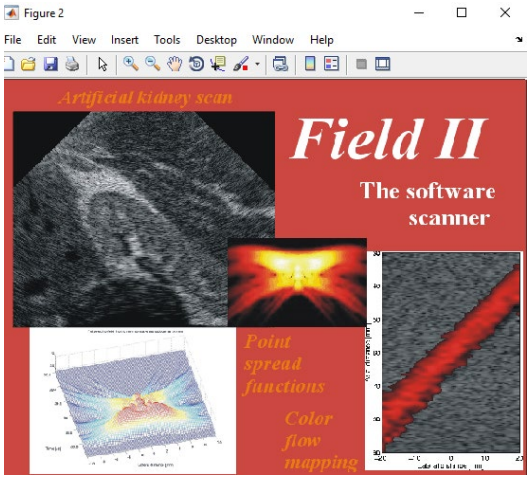


Computer Lab – Imaging setup




```
f0=3e6; % Transducer center frequency [Hz]
fs=100e6; % Sampling frequency [Hz]
c=1540; % Speed of sound [m/s]
lambda=c/f0; % Wave length [m]
width=lambda; % Width of element
element_height=5/1000; % Height of element [m]
kerf=width/20; % Kerf [m]
focus=[0 0 50]/1000; % Fixed focal point [m]
N_elements=192; % Number of elements in the transducer
N_active=64; % Active transducer elements (aperture)
no_lines = 129 % Number of scan lines
```

Lab assignments



- Download Field II Folder (Canvas)
- Initiate Field II (field_init.m)
- Generate the cyst phantom

[positions, amp] = cyst_phantom(N).

N= 5000 number of point scatterers

- Simulate ultrasound image **sim_cyst_phantom.m**.
- Assignments:
  - Different scatter densities
  - Imaging focus depth
  - Phantom modification (e.g.: add a specular reflection, add a second cyst)

Lab report

- Written individually – submitted on Canvas
- Do not have to account for all lab tasks – only reply to following questions:

1) Briefly explain the basic theory of ultrasound image simulation
2) For what purpose can ultrasound image simulations be useful? Please give some examples.
3) Describe the dependence of point scatterer density in the simulations. Illustrate with example images from the lab. How many scatterers are needed in the phantom to simulate fully developed speckles?
4) Describe the difference between a specular reflection and backscattering. How do they show up in the image?
5) Describe how changing the focus depth affected the simulations. Illustrate using example images from the lab.
6) Account for your modifications in Lab assignment 8. Were you able to verify them in simulations?